30-Day AI Challenge Roadmap

ROADMAP A — Generic Version (no context, what a vague prompt produces)

This is what you'd get if someone just typed *"give me a 30-day AI learning roadmap"* with no background info.

Week	Focus	Resource	Project
1	What is AI/ML/DL; Python for data science (numpy, pandas)	Generic "AI for Beginners" YouTube playlist	Python practice exercises
2	ML basics: regression, classification, scikit-learn	Andrew Ng's ML course	Titanic survival prediction
3	Deep learning basics: neural nets, CNNs, TensorFlow/Keras	deeplearning.ai intro videos	MNIST digit classifier
4	NLP basics, chatbots, resume tips	NLP intro playlist	Simple rule-based chatbot

Final outcome: Basic conceptual understanding of AI/ML, a couple of tutorial-style projects, generic resume advice.

This treats the learner as a complete beginner, re-teaches Python/data basics you already know, ignores your time budget, ignores the SWE-vs-AI/ML dual goal, and never mentions prompt engineering, AI tools, or internship application steps — even though your original ask explicitly named all of those.

ROADMAP B — Personalized Version (built from your context)

You: 3rd-year B.Tech IT, strong in C++/Java/Python/SQL/DSA, intermediate level, 2-3 hrs/day, goal = SWE + AI/ML internships, learn by projects + videos.

Week 1 (Days 1-7): AI Tools & Prompt Engineering Mastery

Milestone: You can use AI tools fluently and call an LLM API from your own code.

Day	Task	Resource
1	Map the AI/ML/DL landscape (30 min); set up OpenAI, Claude, Hugging Face accounts	DeepLearning.AI "AI For Everyone" overview video
2	Learn prompt engineering fundamentals: zero-shot, few-shot, chain-of-thought	docs.claude.com prompt engineering guide
3	Practice writing/refining prompts; build a personal "prompt library" doc	DeepLearning.AI "ChatGPT Prompt Engineering for Developers" (free, video)
4	Explore AI coding assistants (Claude/Copilot) on a real coding problem	Hands-on, no external resource needed
5	Learn to call an LLM API with Python (`requests`/SDK)	Anthropic/OpenAI API quickstart docs
6	Build: a CLI tool that summarizes text or answers questions using the API	Project day
7	Clean up code, write a README, push to GitHub	Git basics video if needed

Week 1 Project: AI-powered CLI assistant (your first GitHub repo for this challenge).

Week 2 (Days 8-14): Applied Python for ML

Milestone: You can build and evaluate a basic ML model end-to-end.

Day	Task	Resource
8	NumPy + Pandas crash course (you know Python, so this is fast)	Kaggle "Pandas" micro-course (free)
9	Data visualization: matplotlib/seaborn	CodeBasics YouTube
10	ML concepts: supervised vs unsupervised, train/test split, overfitting	Kaggle "Intro to Machine Learning"
11	scikit-learn regression walkthrough	StatQuest YouTube
12	scikit-learn classification walkthrough	Krish Naik YouTube
13	Start project: pick a Kaggle dataset, build a predictive model	Kaggle datasets
14	Finish project, document, push to GitHub	—

Week 2 Project: End-to-end ML prediction project (e.g., student performance or house price predictor).

Week 3 (Days 15-21): Deep Learning + Real AI App

Milestone: You've built and deployed an actual AI application — your strongest resume piece.

Day	Task	Resource
15	Neural networks intuition (no heavy math first)	3Blue1Brown "Neural Networks" series
16	Build a simple NN on a toy dataset in PyTorch/TensorFlow	freeCodeCamp PyTorch crash course
17	Learn LLM concepts: embeddings, vector databases, RAG	LangChain docs intro
18	Build a RAG app: chat with a PDF/document using embeddings + API	Project day
19	Add a simple UI with Streamlit	Streamlit docs
20	Test, debug, refine the app	—
21	Deploy on Streamlit Community Cloud / Hugging Face Spaces, push to GitHub	—

Week 3 Project: Deployed RAG chatbot ("chat with your notes/PDF") — a genuinely strong AI/ML internship portfolio piece.

Week 4 (Days 22-30): SWE Polish, Portfolio & Internship Push

Milestone: You're an applied candidate, not just a learner — resume, GitHub, and applications are live.

Day	Task	Resource
22	Git/GitHub deep dive: branches, PRs, clean commit history	freeCodeCamp Git course
23	Rewrite resume around your 3 projects; use an AI tool to tighten wording	—
24	Optimize LinkedIn + GitHub profile (pinned repos, profile README)	—
25	DSA refresh (you're already strong — 2 problems/day to stay sharp) + behavioral interview prep	LeetCode, "Tech Interview Handbook" repo
26	System design basics: APIs, databases, scalability fundamentals	Gaurav Sen YouTube
27	Mock technical interview — use Claude/ChatGPT as a practice interviewer	—
28	Polish capstone (Week 3 project): demo GIF, clear README, short LinkedIn post about it	—
29	Shortlist companies/roles, customize applications, start applying	—
30	Review all 3 projects, reflect on gaps, plan Days 31-60	—

Week 4 Output: Polished resume + GitHub + LinkedIn, applications submitted, interview practice done.

Final Outcome After Day 30

- 3 deployed, documented GitHub projects: AI CLI tool, ML predictor, RAG chatbot
- Practical fluency in prompt engineering and AI tool usage
- Working knowledge of applied ML and basic deep learning
- A resume and online presence built around real, demonstrable work
- Active internship applications in flight, with a plan for the next 30 days